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Unit 1 : Project Overview and ~~Feasibility Study~~ Project Scheduling

* Project Overview

Q. what is the concept of Proj overview ? Project = Software

Project — Compute Summary system = Software
- Prj = ? what is project.
- Cause = ? reason behind proj.
- Benefit = ? fayda after proj

i) Prj Name
ii) Prj Description detail
iii) Objectives reason behind proj.
iv) Scope users of the proj.
v) Expected outcome

* ~~Feasibility~~ Study

Proj - ready - - possible ✓
- profit ✗

is it possible to create, will it give profit or not.

There are four types of feasibility study

① Technical Feasibility (is technology available)
proj - Technology
H/W ?
S/W ?

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Q. what is feasability study, state

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Internet speed?
proj. language ?
Database ?

If verifies that, are we technically ready to make the project

② Economic feasibility (will it be profitable or not)
proj - profit will it make profit
proj: cost - ask what is the cost, is it ok
profit - ↑

③ Operational feasibility (will the user use it)
proj - user - system use ✓
- staff - work ✓
- staff - training ✓
- users - happy ✓

④ Schedule feasibility (will it complete in decided time)
Proj - time
ex. Decide - 3m
Complete - 3m ✓

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feasibility study
Ex. College ERP - Budget - ?
Developer = ?
Time = ?
s/w create = ✓
users

* Identification

what is identification

Problem = ?
s/w x

Steps
1. Identify the problem
2. Analyse current situation Accuracy ↑↓
3. Identify stakeholders (users)
4. Gather requirements
5. Define project objective
6. Identify possible solution
7. Preliminary feasibility check (feasibility check)
8. Prepare Project Proposal.

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① Explain the concept of market with ex. ?

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* Market and Demand Analysis

Before starting IT proj → ① Studying the market ② customers behavior ③ future demand for a product.

- Market Analysis

Overall market environment - study
Sub points →
- Market growth
- Industry trends
- Target customers
- Pricing strategy

Eg:
Food delivery app - · No. of online users.
· Market trends
· Food delivery services

- Demand Analysis

How many customers? → will Buy = ? / will use = ? } product

Sub points
- Customer needs
- Product Price
- Population
- Seasonal demand
- Income level (Affordable for customer)

Eg. online learning platform - develop
- No. of students
- Internet availability
- Demand for online course

* Steps to perform Market and demand Analysis

1. Identify the target market (customers)
   - who will use the product = ?
2. Collect Market Information
   - Surveys, interview, online search, etc
3. Analyze customer needs
   - customer expectations
   - Pain points (points the user don't want)
   - Desired features
4. Analyze competitors
   - product offered = ? (by competitors)
   - pricing = ? (their pricing)
   - weakness = ? (their weaknesses)

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5. Estimate Market Demand
- No. of potential customer (genuin customer)
- Future growth = ?
- Expected sales = ?

6. Forecast future demand (prediction of demand)
- Product → future demand
- Historical data
- F/B of customers

7. Prepare Report
- Market size = ?
- Target customers
- Risks
- Business opportunities

Q. Explain the concept of project cost estimate along with components & objectives. 5mm

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* Project Cost Estimate

Process of → predicting the total cost required
↓
total complete an IT project

- Cost of resources
- labor
- H/w, s/w cost
- Training
- Testing
- Maintenance

overall Objective - project Budget

- Objective

1. Determine the total project budget
2. Supports ~~plan~~ project planning & scheduling
3. Allocate resources
4. Reduce financial risk

- Components of proj cost Estimate

1. Human resource cost
   - Project manage
   - s/w dev
   - Tester
   - ui/ux designer

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- Business Analyst

2. H/w cost
   - cost of physical equipments
   - Computer
   - Servers
   - N/w devices
   - Printer

3. S/w cost
   - Expenses related to s/w licensed
   - os
   - Database
   - Antiviruses

4. Training cost
   - Training cost emp on new technologies
   - Workshops
   - Instructor fees
   - online certifications

5. Infrastructure cost
   - Internet
   - electricity
   - office rent
   - furniture.

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* Financial Appraisal
  study of finance of a s/w befor creating it
  → Process - check, - s/w - Profit: ✓
  - feasible: ✓

  → check - proj reason - evaluate

• Objectives
1. To estimate the total project
2. To estimate profit
3. To compare diff. investments
4. To minimize financial risk

• Components
1. Initial Investment (Before s/w build)
   Amt of money → start the proj.
   → H/w cost
   → s/w cost
   → office steup
   → Emp. Training.

2. Operating cost. (During Building Process)
   → Repeated expense - during proj.

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→ Emp. salary
→ Electricity bill
→ Maintenance
→ Internet Charge

3. Revenue estimation

Expected income → generated from proj.

→ Product sales
→ service charges

4. Cash flow Analysis

→ Movement of money IN/OUT
→ Cash Inflow (Income)
→ Cash Outflow (Expense)

• Financial Apparasal Techniques

1. Payback Period

$$PBP = \frac{\text{Initial Investment}}{\text{Annual cash Inflow}}$$

2. Return on Investment (ROI) it find profit in %

$$ROI = \frac{\text{Net Profit}}{\text{Investment}} * 100$$

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3. Benefit Cost Ratio

$$BCR = \frac{\text{Percent value of Benefit}}{\text{Percent Value of Cost}}$$

• Advantages
Also objects can be written here.
1. Helps in Investment decision
2. Measure Profit
3. Reduce risk (financial)

* Project Scheduling

Process → Creating TT for proj activities
- sequences of Tasks
- duration = ?
- assigning resources = ?
- Decision of start & end date

Objectives
1. Planned deadline - Project complete
2. Resources proper use
3. Project progress monitor
4. Project risk *

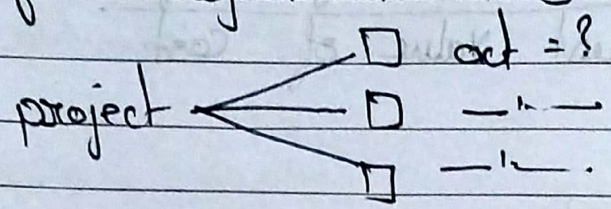
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CPM - Critical Path method

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* Steps in project scheduling

1. Define Project activities



2. Sequence activities - Determine order
3. Estimate activity duration
   Activity - perform = Time?
4. Assign Resources
   - Allocate Team members
   - H/W
   - S/W
5. Develop the project schedule
   - Schedule - Gant chart, Pert, CPM
6. Maintain and update
   - Track
   - Progress - regularly

• Benifits

1. Timely Completion of project
2. Better monitoring
3. Improves project planning

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* PERT Calculation

Variance : Risk factor

Activity	optimistic (to)	most likely (tm)	Pessimistic (tp)	te	σ^2
1-2	1	7	13	7	4
1-6	2	5	14	6	4
2-3	2	14	26	14	16
2-4	2	5	8	5	1
3-5	7	10	19	11	4
4-5	3	5	17	7	4
6-7	5	8	29	11	16
5-8	3	3	9	4	1
7-8	8	17	32	18	16

- Formula

Expected time:

$$te = \frac{to + 4tm + tp}{6}$$

Variance :

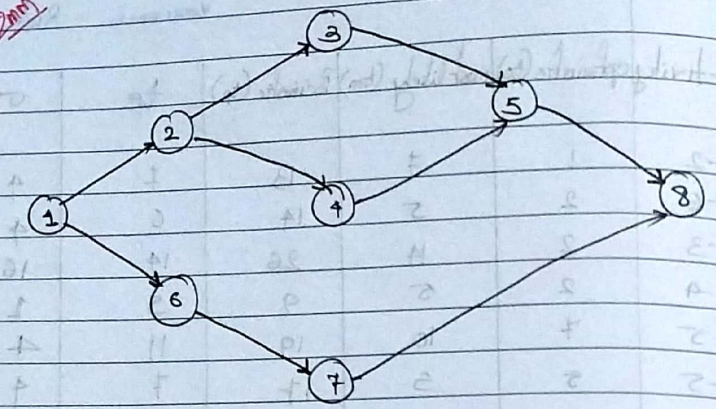
$$\sigma = \left(\frac{tp - to}{6}\right)^2$$

- to calculate
1. Draw project n/w diagram
2. Expected duration and Variance
3. Earliest and latest occurance.
4. Expected project length.

Note: Draw hote table with te, & σ colums after step (2)

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1. Network diagram



2. ① 1-2

$$te = \frac{to + 4tm + tp}{6}$$

$$= \frac{1 + 4(7) + 13}{6}$$

$$= \frac{42}{6}$$

$$= 7$$

$$\sigma = \left(\frac{tp - to}{6}\right)^2$$

$$= \left(\frac{13 - 1}{6}\right)^2$$

$$= 2^2$$

$$= 4$$

② 1-6

$$te = \frac{to + 4tm + tp}{6}$$

$$= \frac{36}{6} = 6$$

$$\sigma = (12/6)^2$$

$$= 4$$

③ 2-3

$$te = \frac{2 + 56 + 26}{6}$$

$$= 84/6$$

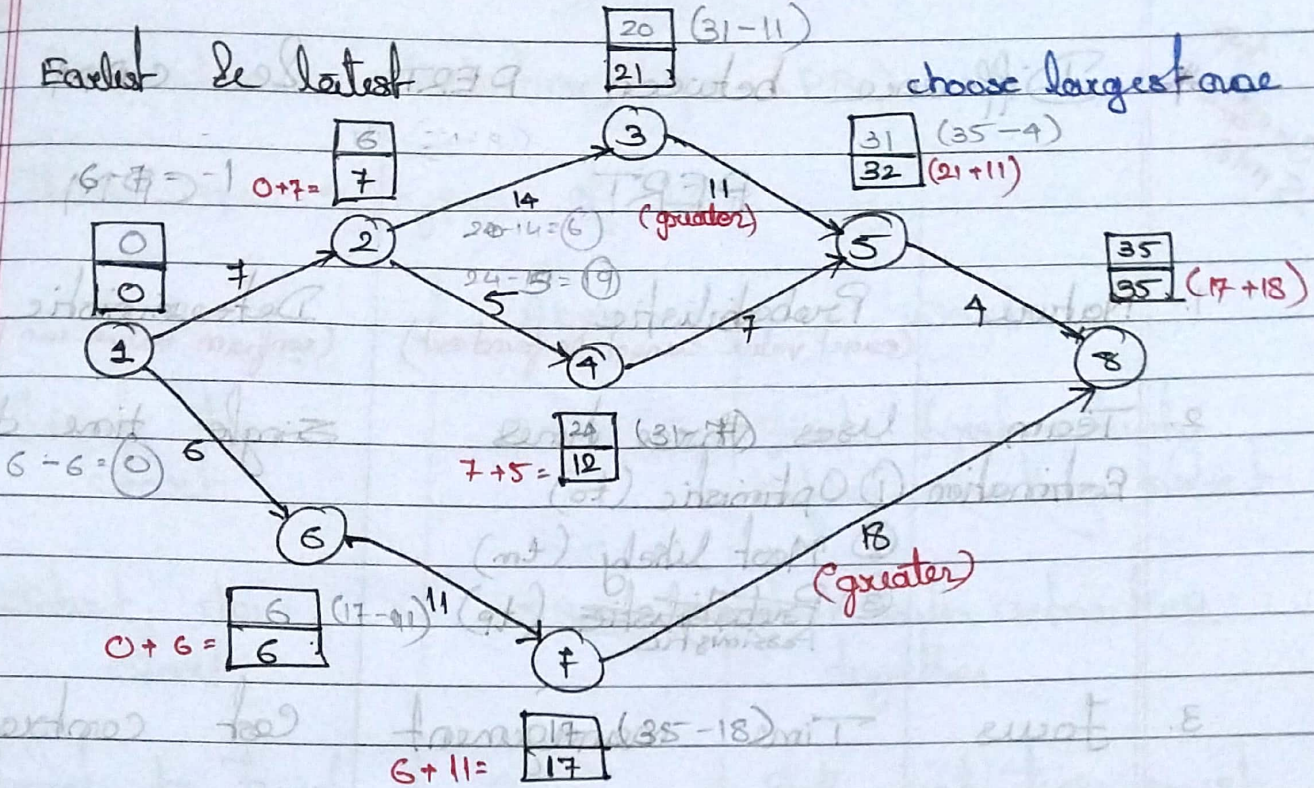
$$= 14$$

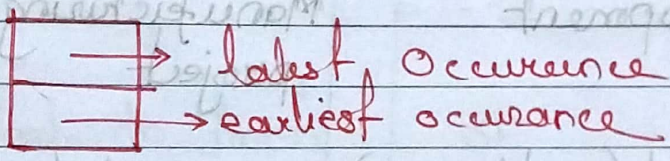
$\sigma =$

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3. Earliest & latest





- Earliest occurance is found using "forward pass"
- leastest occurance is found using "Backward pass"
- Critical path, 1, 6, 7, 8
  If found as 1(0-0=0), 6((6-6=0), 7(17-17=0), 8(35-35=0)

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* Difference between PERT & CPM

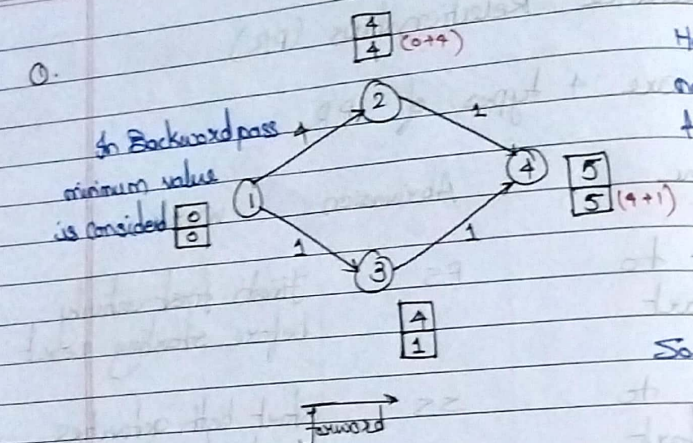
	PERT	CPM
1. Nature	Probabilistic (exact value cannot be found out)	Deterministic (confirm value can be found out)
2. Team Estimation	Uses three times ① Optimistic (t_o) ② Most likely (t_m) ③ ~~Probabilistic~~ Pessimistic (t_p)	Single time duration
3. Focus	Time management	Cost control
4. Suitable for	Development	Manufacturing, repetitive project
5. Orientation	Event (complete combine)	Activity (individual)
6. Cost Consideration	Time → importance cost → no importance	Cost → importance
7. Expected time	$T_e = \frac{t_o + 4t_m + t_p}{6}$	No formula
8. Critical Path	Vary on the situation	Fixed

* Precedance Relationships (PR)

There are 4 types of PR

	Name	Abbreviation	Works
1.	Finish to start	FS	Finish first activity before starting next
2.	Start to start	SS	Start both activities together
3.	Finish to Finish	FF	Both activities must finish together.
4.	Start to Finish	SF	First activity start before second activity finishes.

Backward



Q.

In Backward pass minimum value is considered

Here, to reach node (4) we have two ways i.e from (2) & (3)

from (2) = 4+1=5

(3) = 1+1=2

So we will consider maximum

Therefore, Critical Path

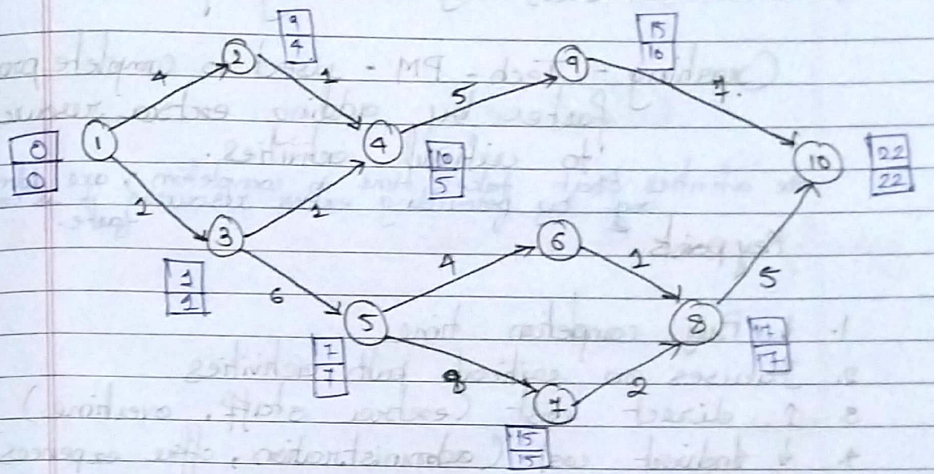
1 - 2 - 4

clear activity

Q. ~~Calculate~~ from the following given question calculate the network diagram, early start timing and late start timing for each activity in order to calculate the critical path.

Network dia
early start / late start
Critical path.

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Duration	4	1	1	1	6	5	4	8	1	2	5	7



Critical path

1 - 3 - 5 - 7 - 8 - 10

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Q. what is crashing? & why it is imp role in PM
How crashing helps in cost reduction

Critical Activity → Activity that mainly takes time

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* Cost reduction by crashing of Activities

Crashing - Tech - PM - used to complete project faster by adding extra resources to critical activities.

the activities that takes time in completion, are identified and by providing extra resources, it is completed faster.

Key points

1. ↓ Proj completion time
2. focuses on critical path activities
3. (↑) direct cost (extra staff, overtime)
4. ↓ Indirect cost (administration, office expenses) (Benefit)

Steps:

1. Identify CP (critical Path)
2. Find activities that can be crashed
3. calculate cost
4. Crash the least expensive critical activity
5. Repeat until the target is achieved

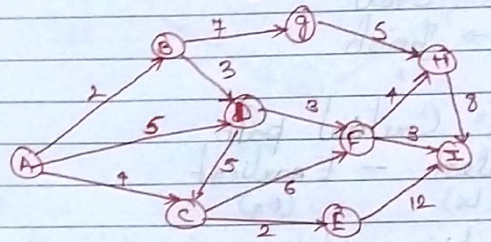
Advantages

1. Finishes the project earlier
2. Reduces overall cost
3. Helps meet deadline

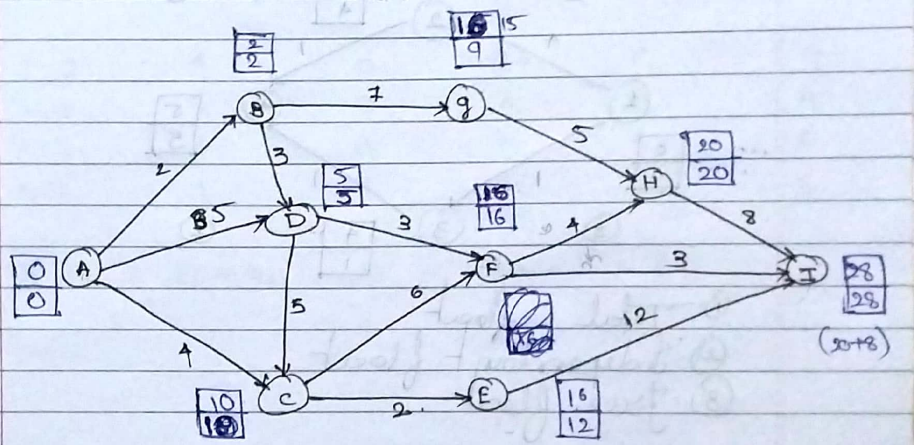
Disadvantages

1. ↑ direct cost
2. Not all activities can be crashed.

Q. Calculate the critical path for the following network diagram



to take min value backward.



to take max value forward

diff btw latest time & earliest

* Float Calculation

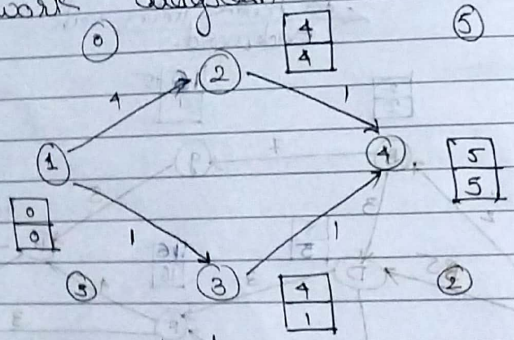
(i) —4→ (j)

T_{ij} = total duration

$E_{S_{ij}}$ - (i) → start
$E_{F_{ij}}$ - (i) → finish
$L_{S_{ij}}$ - (j) → start
$L_{F_{ij}}$ - (j) → finish

Float = ? = Critical path
F = latest (Ls) − Earliest (Es)

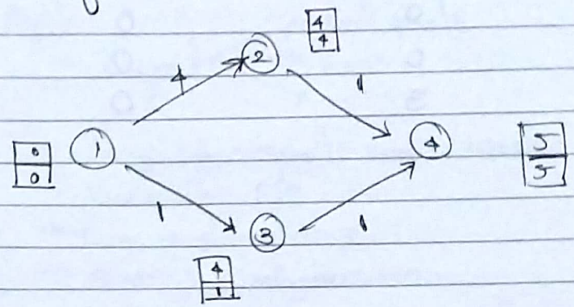
Network diagram



① Total float
② Independant float
③ Free float

Prob

Q. Consider the network diagram (following). calculate forward pass, backward pass. also calculate the float values in the term of total float, independant float, free float



① Activity	② Normal time (t_{ij})	③ start E_S (E_i)	finish E_S (E_i)	start L_S	④ finish L_F (L_j)
1-2	4	0	4	0	4
1-3	1	0	1	3	4
2-4	1	4	5	4	5
3-4	1	1	2	4	5

$E_F = E_i + t_{ij}$ $\quad L_S = L_j - t_{ij}$

P.T.O

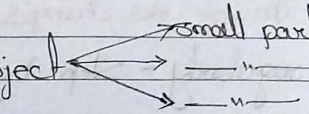
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Total float = $L_s - E_s$	Free float $E_j - E_i - t_{ij}$	Independant float $E_j - L_i - t_{ij}$	E_i (last node)	L_i (last)
0	0	0	4	0
3	0	0	1	0
0	0	0	5	4
3	3	0	5	4

Unit 3: Agile Project Management.

Agile — way of PM (Proj handled in parts)
- step by step implementation
- rather than doing everything at once

In Agile



→ Team - 1 - small proj. - work
→ Customer F/B
→ Team → plan change
→ Team → Work improve

Agile → doing proj in small steps → getting F/B → improving work → proj quality

- Main Benifit

1. Work - faster
2. change - easily apply
3. Customer - F/B - customer interaction → proj.
4. Problem can be found early
5. Team members work together.

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* Agile Principles.

1. Customer satisfaction
2. Welcome changes - Accept changes → proj. proc
   positively accept the changes & implement
3. Deliver work regularly - Step by step → result
4. Work together - Dev, tester, analyst
   group (work)
5. Trust the team - team - function
6. Good Communication
7. Working product is important working ✓ x
8. Keep a steady pace
9. Good quality
10. Keep things Simple
11. Self organizing things
12. Improvement continously

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* Project Audit

- Examination of project → Project
  - purpose
  - effect o/p
  - meet its objectives

Objectives of project Audit

- Check - objectives - achieve (purpose)
- Evaluate cost, schedule (timeline), quality
- Identify pblm and risk
- crosscheck the necessary actions
- Evaluate proj. performance

- Areas covered in project Audit

1. Project planning
   - project - realistic ✓
   - Review budget, schedule, timeline
2. Project Execution
   - Examine actual progress vs planned progress
   - Resources - Property used ✓

3. Cost and Budget
compare actual expenditure vs Approved Budget

4. Quality
standards met ✓ ✗

5. Risk Management
Risk Identify → Action - resolve ✓ ✗

6. Team performance
Evaluate team members ✓ ✗
roles ✓ ✗
co-ordination ✓ ✗

* Project Termination
↓
Process → ending the proj. (Completed) or proj. no longer beneficial (end)

- Types of project termination

1. Normal Termination
- proj - Completed
- objective fullfilled

- Intro para
- types
- steps

} Ans. for proj. termination

2. Premature Termination
- termination before completion
Tech prob, resources, requirement challenge

3. Perpetual Termination
proj - continue → without reaching completion
indefinitely point

4. Failed Termination
Proj - objective unable to fullfill → end
Too expensive
technically impossible
no longer useful

Steps for proj termination

1. Decide to terminate the proj.
2. Review proj status
3. Complete documentation work
4. Release proj resources.
5. Close, contracts, financial account
6. Transfer the proj to customer
7. prepare final report
8. Conduct post proj. review
9. Document lessen learned
10. Formally close the proj.

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* Resource Scheduling

Allocating available proj resources → diff task activities → proj → time, cost, quality

- Types of resources

1. Human Resources → Developers, tasks, proj. mgmt, designer, analyst.
2. H/w resources → N/w devices, Services
3. s/w resources → Database, testing tool
4. financial resources → proj. budget & expenses
5. Time - working hours, proj. duration.

* Steps

1. Identify the activities
   proj → part 1, part 2, part 3, part 4.

short note → Resource scheduling
Resource leveling.

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2. Identify resources
   task → HR, H/w, s/w, cost
3. Estimate Resource
   resources - time req
4. Allocate resources
5. prepare resources schedule
6. Monitor.

* Resource leveling

concept of adjusting the workflow of the s/w → as per → the resource availability

Objectives

1. Eliminate resource over allocation
2. Balance the workload
3. Improve resource utilization

Task A → Develop (mon - wed)
Task B → Develop (Tue - Thur)

TA ≠ TB as here, only 1 dev is there. the we can change the time duration of task

- Advantages

1. Resource overloading prevent
2. Improves productivity
3. Reduces stress in environment

* Project Cost Control

Process of maintaining and controlling proj. expenses — project ← time, app budget

- PERT / cost

Technique → used to find the estimated cost

pecimistic cost (Ep) → ↑ positive cost
optimistic cost (Co) → ↓ probable cost
most likely cost (Cm) → most probable cost

Estimated Cost

$$Ec = \frac{Co + 4(Cm) + (Cp)}{6}$$

- Main Activities

1. Cost estimation →
2. Budget preparation
3. Cost monitoring
4. Corrective Actions.

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